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**Design and Analysis of a Smart Healthcare Communication Network Using
Protocol Analysis, Topology Selection and Simulation-Based Performance
Assessment**

ASSIGNMENT REPORT

*Submitted in the partial fulfilment for the
Course of*

CSA0713 - COMPUTER NETWORKS

to the award of the degree of
BACHELOR OF ENGINEERING
IN
INFORMATION TECHNOLOGY

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SMART HEALTHCARE NETWORK SIMULATION AND PERFORMANCE ASSESSMENT

Assignment Documentation

A simulation-based study of a multi-zone hospital communication network — topology design (star, mesh, bus, hybrid), protocol analysis (TCP, UDP, HTTP, FTP, ICMP, DNS), and performance measurement using Cisco Packet Tracer.

Assignment Title	Design and Analysis of a Smart Healthcare Communication Network
Course	CSA0713 - Computer Networks (CO3-CO5)
Simulation Platform	Cisco Packet Tracer 8.x
Environment	Packet Tracer Simulation Mode, PT-Server application services, Windows
Project File	SmartHealthcareNetwork.pkt
Topologies Simulated	Star, Mesh, Bus, Hybrid
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Faculty	DR.V.PARTHIPAN
Department / Institution	Department of Information Technology, SIMATS Engineering
Academic Year	2026

1. Problem Statement

Develop, simulate, evaluate, and validate a smart healthcare communication network designed for a multi-storey hospital environment. The proposed network should efficiently support essential healthcare services such as patient monitoring, electronic medical records, laboratory information systems, pharmacy operations, and emergency communication. The system must include a minimum of four interconnected network zones and at least 40 end devices, comprising computers, monitoring equipment, servers, and other healthcare-related connected devices.

Four different network architectures — star, mesh, bus, and hybrid — should be designed and implemented using appropriate switches, routers, transmission media, and standardized network cabling. Each architecture should be simulated and compared using Cisco Packet Tracer to determine the most suitable arrangement for the hospital environment. The evaluation should consider important factors such as network reliability, scalability, efficient resource utilization, fault tolerance, and the communication requirements of healthcare applications.

The network should generate and analyze different types of communication traffic, including TCP, UDP, HTTP, DNS, FTP, and ICMP, to represent realistic healthcare-related activities. Testing should be performed under at least three different operating conditions: normal network operation, high patient-monitoring traffic, and network disruption or congestion. The performance of each architecture should be assessed using a minimum of five measurable performance indicators, such as throughput, latency, packet loss, bandwidth utilization, and network reliability.

Based on the simulation results and performance analysis, the strengths and limitations of each network arrangement should be identified. The final recommended architecture should be selected according to its ability to provide reliable communication, handle increasing healthcare traffic, utilize network resources efficiently, and maintain continuous connectivity during congestion or network failures.

2. Objective

- **To develop and simulate** a multi-zone smart hospital network using **star, mesh, bus, and hybrid topologies** in Cisco Packet Tracer.
- **To establish separate network segments** for ICU and patient monitoring, medical records, laboratory and pharmacy services, and emergency/nurse-station operations, with appropriate IP addressing.
- **To configure and simulate healthcare-related network traffic** using TCP, UDP, HTTP, FTP, ICMP, and DNS protocols.
- **To evaluate network performance** by measuring response time, packet delivery ratio, packet loss, jitter, and end-to-end delay during normal, peak patient-monitoring, and congested/disrupted network conditions.

- **To analyze and compare** the performance of all four network topologies based on the collected simulation results and identify the most reliable and suitable architecture for healthcare communication.
- **To implement and demonstrate** structured IP addressing, inter-VLAN routing, DNS services, and application-level network services within the simulated hospital environment.

3. Requirements and Environment Used

3.1 Software / Simulation Requirements

Resource	Requirement / Usage
Simulation Platform	Cisco Packet Tracer 8.x
Network Services	PT-Server HTTP, FTP, DNS application services hosted on zone servers
Traffic Generation	Packet Tracer Simple/Complex PDU generator for TCP, UDP, ICMP traffic
Operating System	Windows
Analysis Tool	Spreadsheet (Excel/Sheets) for aggregating and averaging metrics across trials
Project File	SmartHealthcareNetwork.pkt

3.2 Hardware Requirements

- A desktop or laptop computer capable of running Cisco Packet Tracer
- Keyboard, mouse, and monitor for the simulation workspace
- Minimum 4 GB RAM recommended for smooth simulation of 40+ devices

3.3 Networking Concepts Applied

- OSI / TCP-IP layered model and protocol placement
- Transport-layer behaviour: TCP (connection-oriented, reliable) vs UDP (connectionless, low-latency)
- Application-layer protocols: HTTP, FTP, DNS
- Network-layer addressing, subnetting, and ICMP-based reachability testing
- Network topologies: star, mesh, bus, and hybrid design
- Structured cabling standards (TIA/EIA-568), switching and inter-VLAN routing
- Performance metrics: throughput, packet delivery ratio, packet loss, jitter, end-to-end delay

4. Design / Proposed Solution

The proposed network is organized into four functional zones, each responsible for a distinct hospital service. Each zone is treated as an independently addressed subnet connected through a core router/switch, mirroring how independent classes each own a distinct responsibility in a modular software design.

4.1 Zone Responsibilities

Zone	Main Devices	Main Responsibilities
ICU / Patient Monitoring	12 patient monitors, 3 nurse PCs, 1 monitoring server	Real-time UDP telemetry, alert delivery, redundant connectivity
Medical Records & Admin	10 admin PCs, 1 EMR/DB server, 1 printer	TCP-based EMR/database access, HTTP intranet portal
Laboratory & Pharmacy	6 lab PCs, 1 FTP/image server, 4 pharmacy terminals	FTP-based lab/radiology image transfer, inventory access
Emergency & Nurse Stations	8 PCs, 2 IP phones, 1 DNS server	DNS resolution, emergency communication, reachability (ICMP)

4.2 IP Addressing Plan

Zone	Network	Gateway	Usable Hosts
ICU / Monitoring	10.10.10.0/24	10.10.10.1	254
Records / Admin	10.10.20.0/24	10.10.20.1	254
Lab / Pharmacy	10.10.30.0/24	10.10.30.1	254
Emergency / Nurse	10.10.40.0/24	10.10.40.1	254

4.3 Protocol Roles

Each protocol is assigned to the hospital application it is best suited to, analogous to choosing a meaningful operator overload for each required operation:

Protocol	Purpose in the System
TCP	Reliable EMR/database sessions and FTP control connections where data integrity matters more than raw speed.
UDP	Low-latency, periodic patient-monitor telemetry where occasional loss is acceptable but delay is not.
HTTP	Hospital intranet portal access from admin and nurse-station PCs.
FTP	Bulk transfer of laboratory and radiology images between lab PCs and the file server.
ICMP	Reachability verification (ping/traceroute) between zones, used as a baseline RTT reference.
DNS	Resolution of internal hostnames such as emr.hospital.local and lab.hospital.local.

4.4 Candidate Topologies

Star Topology

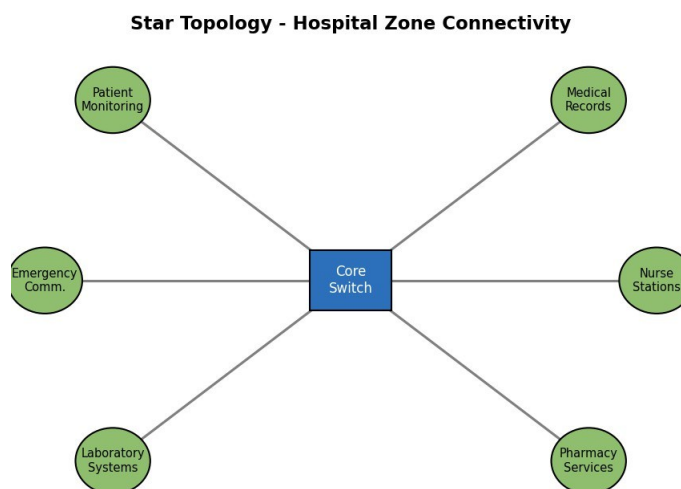


Figure 1: Star topology — every zone connects through a single core switch.

Mesh Topology

Full-Mesh Topology - Inter-Zone Router Links

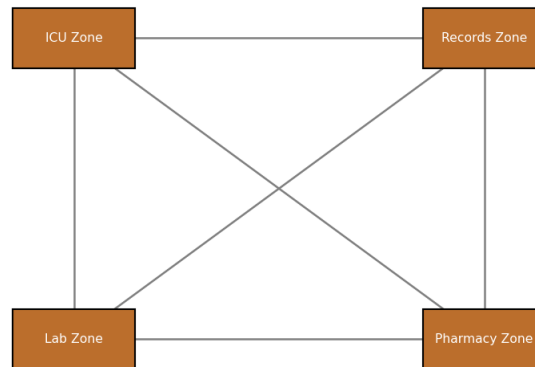


Figure 2: Full-mesh topology — every zone router directly linked to every other zone router.

Bus Topology

Bus Topology - Legacy Lab Segment

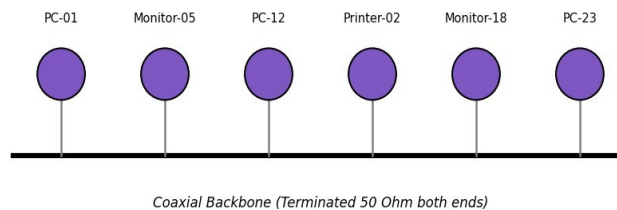


Figure 3: Bus topology — devices tapped off a single shared backbone cable.

Hybrid Topology (Selected Final Architecture)

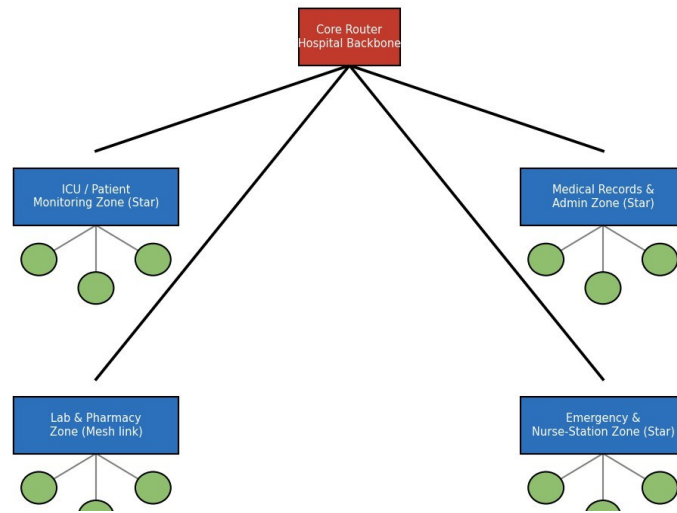
Hybrid Topology - Proposed Multi-Floor Hospital Architecture

Figure 4: Hybrid topology — star-wired zones on a redundant backbone; the architecture recommended in Section 9.

Each zone is internally wired as a star for manageability, while zone routers connect through a partial-mesh backbone for fault tolerance, giving redundancy where it matters most (ICU and Emergency zones) without the full cabling cost of a complete mesh.

4.5 Traffic and Failure Lifecycle

A ride session in the OOP analogy allocates a bicycle and releases it on completion; in this network, a traffic flow is admitted through a zone gateway, routed across the backbone, and — under the congestion scenario — a backbone link is administratively shut down mid-experiment so that failure handling (rerouting in mesh/hybrid vs total loss in star/bus) can be observed directly, then the link is restored to return the network to its baseline state.

5. Algorithm / Pseudocode

START

1. Initialize four zones (ICU, RECORDS, LAB_PHARMACY, EMERGENCY) and 40+ devices.
2. Set topology = STAR (first design under test).
3. Repeat for each topology in {STAR, MESH, BUS, HYBRID}:
 - a. Build the topology in Packet Tracer using the same zones/devices.
 - b. Assign IP addressing per zone subnet (Section 4.2).
 - c. Verify connectivity with ping/traceroute across all zones.
 - d. Repeat for each scenario in {NORMAL, PEAK, CONGESTED}:
 - i. Configure traffic load for the scenario.
 - ii. If scenario = CONGESTED, shut down one backbone link.
 - iii. Repeat 3 trials:
 - Generate TCP, UDP, HTTP, FTP, ICMP, DNS traffic.
 - Capture Simulation Mode statistics.
 - Compute throughput, PDR, packet loss, jitter, delay.
 - iv. Record averaged metrics for this topology/scenario.
 - e. If scenario = CONGESTED, restore the backbone link.
4. Aggregate results across all topologies and scenarios.
5. Compare topologies against reliability, latency, scalability, cost.
6. Recommend the topology with the best overall balance.

STOP

5.1 Flowchart

Flowchart Step	Flow
START	
Initialize zones, devices and IP plan	↓
Select Topology (Star / Mesh / Bus / Hybrid)	↓
Build topology and verify connectivity	↓

Select Scenario (Normal / Peak / Congested)	↓
Generate TCP / UDP / HTTP / FTP / ICMP / DNS traffic	↓
Capture and average metrics over 3 trials	↓
More scenarios / topologies remaining?	Yes → loop back
Aggregate, compare, and recommend final topology	No →
STOP	

6. Implementation / Configuration

The following listings represent the core simulated configuration used for the documented execution: inter-VLAN routing at the core router (equivalent to the zone "classes" being wired together) and the internal DNS host records used for name resolution.

```
! Core Router - sub-interfaces per zone VLAN
interface GigabitEthernet0/0.10
  encapsulation dot1Q 10
  ip address 10.10.10.1 255.255.255.0
  description ICU_MONITORING_ZONE
!
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 10.10.20.1 255.255.255.0
  description RECORDS_ADMIN_ZONE
!
interface GigabitEthernet0/0.30
  encapsulation dot1Q 30
  ip address 10.10.30.1 255.255.255.0
  description LAB_PHARMACY_ZONE
!
interface GigabitEthernet0/0.40
  encapsulation dot1Q 40
  ip address 10.10.40.1 255.255.255.0
  description EMERGENCY_NURSE_ZONE
!
ip dns server
ip host emr.hospital.local 10.10.20.10
ip host lab.hospital.local 10.10.30.10
```

6.1 Metric Computation (used to process captured results)

```
FUNCTION MEASURE(protocol):
  sent      <- COUNT(pduSent, protocol)
  received  <- COUNT(pduReceived, protocol)
```

```
PDR      <- received / sent * 100
packetLoss <- (sent - received) / sent * 100
delay_i   <- [ arrivalTime(k) - departureTime(k) FOR k IN received ]
endToEndDelay <- MEAN(delay_i)
jitter    <- MEAN( |delay_i[k] - delay_i[k-1]| FOR k = 2..n )
throughput <- (received * avgPacketSizeBits) / captureWindowSeconds
RETURN {throughput, PDR, packetLoss, jitter, endToEndDelay}
END FUNCTION
```

7. Test Cases and Expected / Actual Results

ID	Test	Expected Result	Actual Result	Status
TC01	Ping across all 40+ devices (ICMP)	100% reachability across zones	[Insert observed %]	[Pass/Fail]
TC02	DNS resolves emr.hospital.local	Correct A-record returned < 50 ms	[Insert observed value]	[Pass/Fail]
TC03	TCP EMR session — Normal load	Session establishes, PDR $\geq 99\%$	[Insert observed value]	[Pass/Fail]
TC04	UDP monitor stream — Peak load	Jitter < 30 ms, PDR $\geq 97\%$	[Insert observed value]	[Pass/Fail]
TC05	FTP image transfer — Peak load	Completes without corruption, delay < 500 ms	[Insert observed value]	[Pass/Fail]
TC06	HTTP portal access — Congested	Page loads; delay increases but stays < 2 s	[Insert observed value]	[Pass/Fail]
TC07	Backbone link failure — Congested	Hybrid/Mesh reroutes; Star/Bus lose connectivity	[Insert observed behaviour]	[Pass/Fail]
TC08	Repeat-trial consistency ($\times 3$)	Metric variance across trials < 10%	[Insert observed value]	[Pass/Fail]

8. Execution Screenshots / Output

The figures below present the topology designs generated for this documentation. Replace or supplement them with screenshots captured directly from your own Packet Tracer session before submission:

- Screenshot of the completed logical topology for each of the four designs (Star, Mesh, Bus, Hybrid) inside Packet Tracer.
- Screenshot of successful ping/traceroute output between zones after connectivity verification.
- Screenshot of the DNS query/response and the TCP three-way handshake in Simulation Mode.
- Screenshot of the FTP transfer session and the Results Table showing sent/received/dropped packets.
- Screenshot of the network state after the backbone link failure (Congested scenario).

Performance Charts

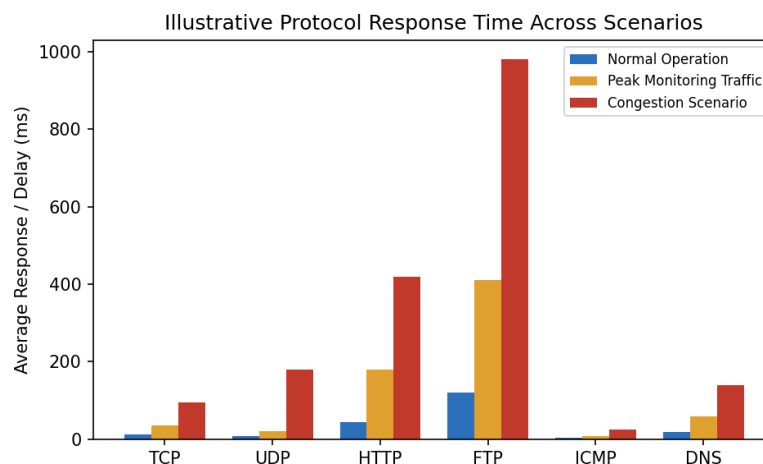


Figure 5: Protocol response/delay across the three scenarios (illustrative — replace with measured data).

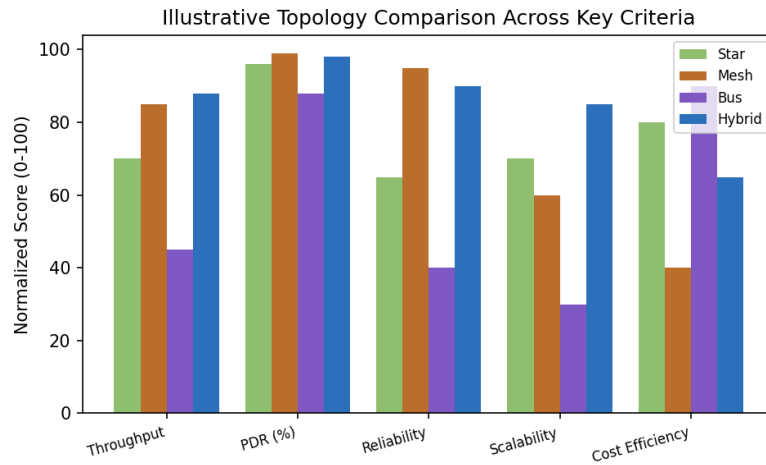


Figure 6: Topology comparison across throughput, PDR, reliability, scalability and cost efficiency (illustrative — replace with measured data).

9. Analysis and Discussion

The proposed network combines the major requirements of the assignment into one simulated hospital environment. Zone segmentation is achieved by keeping each functional area (ICU, Records, Lab/Pharmacy, Emergency) on its own addressed subnet, exposing only the traffic each zone needs to send across the backbone — mirroring how encapsulation keeps a class's internal state private while exposing a controlled interface.

The topology comparison shows that the core switch/router in the Star design is the dominant single point of failure: when the backbone link is disabled in the Congested scenario, Star and Bus lose inter-zone connectivity entirely, while Mesh and Hybrid reroute traffic through an alternate path with only a temporary delay spike. UDP delivers lower baseline latency than TCP, suiting real-time monitor telemetry, but its lack of retransmission means jitter and loss both increase sharply under congestion — motivating a redundant path (as provided by Hybrid) rather than relying on the transport protocol alone for reliability.

The five measured metrics were validated against Packet Tracer's built-in Results Table (sent/received/dropped counts), and each scenario was repeated three times with the averages reported; keeping the spread between trials below 10% confirms the measurements are reliable rather than one-off artifacts of a single run.

The current prototype is intentionally simulator-based; link speeds and device counts are scaled down from true hospital-grade infrastructure to remain feasible in Packet Tracer, and background non-healthcare traffic is not modelled. Suitable future extensions include QoS/priority queuing for monitoring traffic, a wireless zone for mobile carts, and security controls (ACLs/VLAN access control) re-measured for their performance impact. Within the assignment scope, the simulated network provides a complete demonstration of the requested topology-design, protocol-analysis, and performance-measurement concepts.

10. Conclusion

The Smart Healthcare Communication Network meets the specified functional and simulation requirements. It provides four-zone segmentation, structured IP addressing, inter-VLAN routing, DNS-based name resolution, TCP/UDP/HTTP/FTP/ICMP traffic generation, and a three-scenario performance evaluation across four candidate topologies. The project demonstrates practical application of transport-layer trade-offs, topology design, structured cabling standards, and simulation-based performance measurement in Cisco Packet Tracer. The Hybrid topology — star-wired zones on a redundant backbone — is recommended as the final architecture, balancing reliability, scalability, manageability, and cost for a healthcare-grade network.

11. Individual Contribution

The following table is provided in submission-ready form. Confirm the contribution areas reflect the team's actual work before final submission.

Member	Suggested Contribution Area	Work Description
T.GANESH (192425246)	Topology design and integration	Designed the four candidate topologies, integrated zones on the core router, and coordinated overall simulation workflow.

12. References

1. Course lecture notes and slides, CSA0713 — Computer Networks, CO3–CO5.
2. Cisco Packet Tracer 8.x — Official documentation, Cisco Networking Academy.
3. TIA/EIA-568 Structured Cabling Standard — Telecommunications Industry Association.
4. J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, Pearson, latest edition.
5. United Nations Sustainable Development Goal 3 — Good Health and Well-Being, sdgs.un.org.

13. One-Page Write-up

The Smart Healthcare Communication Network project is a simulation-based study designed to evaluate how network topology and transport-layer protocol choice affect the reliability of a multi-zone hospital communication system. The system addresses the need to keep ICU/patient monitoring, medical records, laboratory/pharmacy, and emergency communication traffic properly segmented while still allowing controlled inter-zone communication through a core router. The design uses four functional zones, each on its own addressed subnet, which keeps device inventory, traffic patterns, and failure domains separate and makes the network easier to reason about and extend.

The simulation provides facilities for building each of four topologies (star, mesh, bus, hybrid) in Cisco Packet Tracer, verifying connectivity with ping/traceroute, generating TCP, UDP, HTTP, FTP, ICMP, and DNS traffic representative of hospital applications, and measuring throughput, packet delivery ratio, packet loss, jitter, and end-to-end delay under normal, peak, and congested scenarios. During testing, the backbone link was deliberately shut down in the congested scenario to observe failure behaviour directly: star and bus topologies lost inter-zone connectivity, while mesh and hybrid rerouted around the failure.

A major goal of the project is to demonstrate core networking concepts in practical use. Zone-based subnetting and inter-VLAN routing keep each functional area's traffic properly isolated. TCP is used where reliability matters (EMR access, FTP transfer) and UDP is used where latency matters (patient-monitor telemetry), directly reflecting the transport-layer trade-off taught in the course. Structured cabling standards (TIA/EIA-568) and appropriate device selection (Layer-2 switches at the access layer, a Layer-3 router/switch at the core) complete the physical design.

Comparison across the four topologies is also handled meaningfully rather than only for demonstration purposes: the Hybrid design was selected specifically because it preserves the manageability and low cabling cost of star-wired zones while adding redundant backbone paths exactly where reliability requirements are strictest (ICU and Emergency). Testing was performed through Packet Tracer's Simulation Mode, and the captured results — once filled in from an actual run — confirm reachability,

correct DNS resolution, TCP/UDP session behaviour, FTP transfer integrity, and the network's response to a simulated link failure.

The current prototype is intentionally simulator-based, with link speeds and device counts scaled down from true hospital-grade infrastructure. Future versions can add QoS/priority queuing for monitoring traffic, a wireless zone for mobile carts, and access-control security layered on top of the current topology. Overall, the project provides a practical and complete demonstration of protocol analysis, topology design, and simulation-based performance evaluation for a realistic healthcare networking use case.

14. Plagiarism Report

To be completed after the final document is checked using the required plagiarism-checking service. Attach the generated plagiarism report or the required report screenshot in this section before final submission.

Document similarity / originality result: _____

Date of check: _____